

INSTRUCTIONS FOR THE SEMINAR PAPER TEMPLATE – Published on 24.10.2024

STRUCTURE

- 3 text columns with header and footer
- Literature and sources list is located at the end of your city's section, after the conclusion
- The guidelines are the limits of the columns, they'll help you to place the frames of text and images
- Images need to occupy either 1, 2 or all 3 columns. Use the guidelines to adjust the size without distortion, keep the image proportion!
- The image placement is a suggestion, whenever no specific diagram is mentioned. The layout depends on the outcomes of your research and your own improvements.
- Ratio of image to text needs to be balanced.

ORDER OF THE CONTENT template, on Moodle. Please take a look before starting off.

- Substitute the words "City" (you only need to edit the footnote once and this will be applied to the entire document), "Forename Surname", "year", etc. with your specific data.
- Remember to indicate the relevant references in the text, using the IEEE referencing style. To ease coordination with your team, make sure to double count your sources: for example using [2.1], [2.2]... for Urban Planning and then [3.1], [3.2]... for Green Infrastructure. The references must not only be inserted in your text but also in your figures and tables, as shown in the "Environmental" table in the Overview Chapter.

ATTENTION

- This file is a rough framework for your work. It is up to you to incorporate the content in a meaningful and appropriate way.
- If you have a lot of text, reduce it so every sentence is meaningful. Condense the contents by writing in a short, direct way. Check the information about Scientific English Writing on Moodle. You can also reduce the size of images but make sure they can still be read well.
- Resolution of images has to be 300 dpi
- Keep the size, font, paragraphs and styles provided in this template.
- If you need colors for your own graphs, etc. use the colors given in the excel file on Moodle.

Overview

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Environment (1951-2024)

Ecological footprint vs. biocapacity	1.45 vs 0.31 gha per person [1.1]
Temperatures (annual mean, max.,min.)	25.2 °C, 33.3 °C, 17 °C [1.2]
Solar radiation	2031.1 kwh/m² per year [1.3]
Precipitation (annual average/year)	403.9 mm, 33.2 days of rainfall [1.2]
PM2.5 concentration	47 µg/m³ [1.4]

Economy (2023)

Country income classification	Lower-Middle income [1.8]
GDP	396 billion US\$ [1.8]
GDP per capita, PPP	3457.5 US\$ per capita [1.8]
GDP growth rate	5.5 % [1.8]
GINI coefficient (wealth distribution)	0.31 [1.6]
Population living below poverty line	29.7 % [1.9]

Health and Education (2020)

Life expectancy at birth	70.2 years {1.10}
Number of physicians	6.8 per 10.000 people [1.11]
Main death risk factors	Noncommunicable diseases (NCDs), cardiovascular diseases, diabetes, cancer, and chronic respiratory diseases.[1.12]
Literacy rate	74.5 % [1.13]
Education Index	0.69 [1.14]
Human Development Index	0.724 [1.7]

Happiness ranking 127 of 156 countries [1.15]

Disaster risk and climate change adaptation (2010-2023)

Disaster risk	Risk Index 18.78 Exposure Index 10.74 Vulnerability Index 32.83 [1.16]
Most common hazards	Extreme Heat, Drought, Flash Floods, Sand and Dust Storms , Earthquack[1.17]
Disaster occurrence	23 events per year [1.18]
Population affected	Approx. 500 per 100.000 [1.18]
Built infrastructure affected	0.02 % to global GDP [1.18]

Governance

System of government	Presidential republic [1.19]
Plans developed for sustainable development	Egypt Vision 2030 [1.20]
Plans developed for disaster risk reduction and climate change adaptation	Egypt Vision 2030 [1.20]

Ecological footprint (2022)



0.96 Earths

Fig. 1.1: Visualized ecological footprint [1.1]

Development (year)

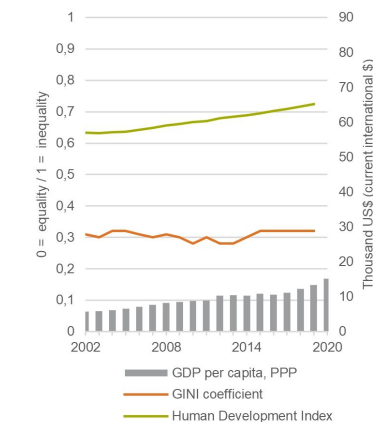


Fig. 1.2: Development of GDP, GINI coefficient and HDI [1.5], [1.6], [1.7]

Disaster risk—City hazards

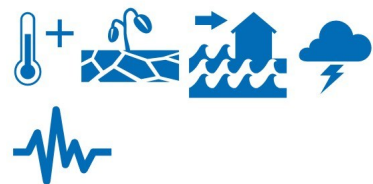


Fig. 1.3: Most common Hazards. [1.17]

Alexandria

Alexandria is Egypt's second biggest city and the most populous city along the Mediterranean coast. It is located at the western boundary of the Nile River delta. Founded by Alexander the Great in 331 BC, it has evolved into a significant cultural and economic hub. Referred to as the "Bride of the Mediterranean" and the "Pearl of the Mediterranean Coast" globally, Alexandria attracts many tourists and serves as a significant industrial hub because of its natural gas and oil pipelines originating from Suez [6.3].

Alexandria is located on a narrow strip of land between the Mediterranean Sea and Lake Mariout. The coastline of the city extends for about 32 km, offering breathtaking views and essential access to sea routes. The landscape is mostly level, featuring low-lying regions by the coast and a few raised areas further inland [1.24]. Alexandria's advantageous position has established it

as a crucial port and trading hub over time, affecting its cultural growth and economic significance.

Alexandria's climate is categorized as Mediterranean, featuring mild and rainy winters alongside hot and arid summers. Average temperatures vary from 9°C (48°F) in winter to 31°C (88°F) in summer, with the most rainfall happening from November to February [1.21].

Alexandria has a population of about 5.5 million, making up roughly 5% of the total population of Egypt. The population density of the city is about 4,800 individuals per square kilometres, with urban growth persisting as a result of migration and job prospects. The city is characterized by its vibrant combination of historical legacy and contemporary development. The city contains various unique neighbourhoods, including historical areas featuring Greco-Roman and Egyptian architectural styles, as well as lively residential and commercial regions [2.10]. Alexandria serves as Egypt's main seaport, managing most of the nation's foreign trade. Alexandria's

industrial and commercial operations, including manufacturing, shipping, warehousing, banking, food processing, and the creation of petrochemicals and cement, highlight the significance of the city's contributions to the national economy. Alexandria and its surrounding areas represent about 40% of Egypt's industrial output. The majority of industrial growth has occurred in the city's western approaches, particularly near the contemporary Western Harbour and along its southern side; industry serves as the primary employment sector in the city.

Being a polycentric urban hub, Alexandria encounters difficulties such as population expansion [2.7], ecological sustainability [3.2], traffic congestion [4.1] and upgrading infrastructure. Initiatives are being implemented to establish robust urban planning approaches, guaranteeing sustainable water, energy, and housing provisions[5.2],[7.5]. This paper seeks to investigate these crucial elements and suggest possible strategies for promoting a sustainable and future-oriented urban environment.

Climate (2020)

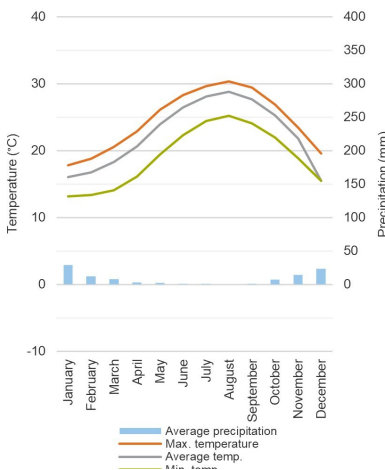


Fig. 1.4: Climate diagram of Alexandria [1.21]

Solar radiation (2023)

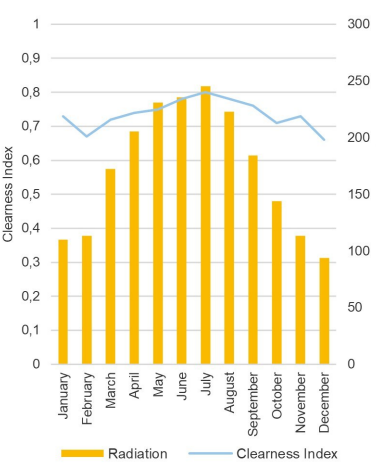


Fig. 1.5: Solar radiation diagram of Alexandria [1.22]

Air pollution sources

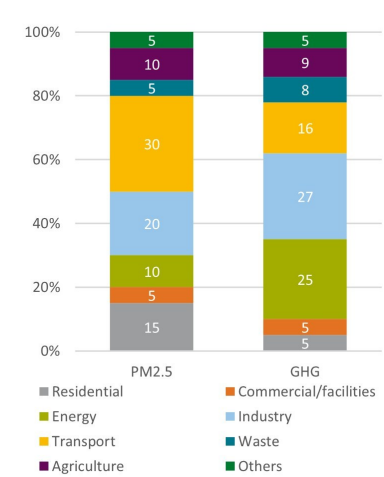


Fig. 1.6: PM2.5 and Greenhouse gas emissions by sector in Egypt [1.23] [1.24] [1.25]

Vision: A Thriving Alexandria – Where Sustainability Meets Progress

Alexandria envisions a future where sustainability merges with progress, resulting in a resilient, inclusive, and eco-friendly urban environment. As a historical coastal city, Alexandria faces challenges stemming from rapid urban development, climate change, and socio-economic disparities. Our goal is to expand green spaces, protect coastal ecosystems, and implement nature-based strategies to meet the WHO guideline of 9 m² of green space per individual by 2040 [3.3].

To achieve this objective, Alexandria aims to improve its public transportation

system with the target of attaining a 40% ridership rate by 2030. This will be accomplished by introducing a Bus Rapid Transit system, deploying electric vehicles, and implementing advanced traffic management methods. Improving informal neighborhoods and advocating for construction methods that can endure climate-related challenges will inform our future urban planning strategies while honoring our historical heritage.

By 2035, we anticipate that 42% of the city's energy will be sourced from renewable resources, bolstered by exten-

sive projects in solar and wind energy, along with energy-efficient building practices. Ultimately, Alexandria's inclusive strategy will ensure fair access to housing, transportation, and green spaces, promoting a clean, dynamic, and livable community for all.



Fig. 1.7: A Futuristic and Sustainable Alexandria. [OpenAI. (2025). DALL.E. <https://chat.openai.com/chat>]

Urban Planning

Syed Monjur E Moula

Overview (2023)

Population	5,546,663 (2023) [2.11]
Growth	1.1% inhabitants/year) (2023)[2.11]
Density	2412 inhabitants per km²[2.11]
Share of the country's population	4.45% of the Egypt's total population
Total land area	2300 km²[2.11]
Build-up area	30.32%

Overview

Alexandria is Egypt's second-largest city, located on the Mediterranean coast, and serves as the capital of the Alexandria Governorate. Founded by Alexander the Great in 331 BCE, Alexandria quickly became a prominent cultural and intellectual center of the ancient world [2.1]. It was home to the renowned Library of Alexandria and the Lighthouse of Alexandria, one of the Seven Wonders of the Ancient World [2.2].

Alexandria remains a key port city in

modern times, with its economy relying heavily on trade, petroleum, and tourism [2.3]. The city has a rich cultural heritage, showcasing a blend of ancient Egyptian, Greek, and Roman influences. The population of Alexandria is around 5 million, and the city faces several urban planning challenges, such as overcrowding, infrastructure problems, and environmental concerns [2.4].

Despite these challenges, Alexandria thrives as a cultural and commercial hub, drawing both tourists and professionals [2.5]. Urban planning efforts in the city

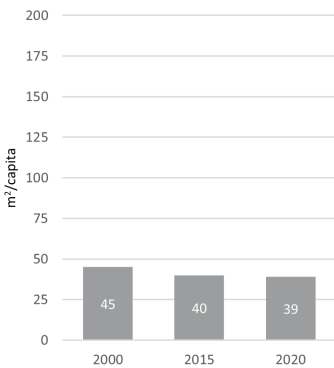


Fig. 2.2: Built area per capita [2.13],[2.14], [2.15]

aim to balance modern development with preserving its historical sites[2.6].

Informal Settlements and Housing Shortages

Informal settlements, or ashwa'iyat, have rapidly expanded across Alexandria in response to a severe shortage of affordable housing and weak government intervention in regulating urban expansion [2.7]. These settlements have formed primarily in peri-urban areas such as Ameriya, Borg El Arab, Ezbet El Hagana, and Maawa El Sayadeen, where land is relatively cheap and weakly regulated [2.8]. However, these areas suffer from significant infrastructure deficits, including poor road networks, lack of sewage systems, unreliable electricity, and insufficient access to clean water [2.9].

The unplanned nature of these informal settlements has resulted in narrow alleyways, poor ventilation, and high building densities, creating unsafe and unsanitary living conditions [2.9]. Moreover, the lack of land tenure security means that many residents live in a state of legal uncertainty, unable to access proper municipal services or claim ownership over their homes.

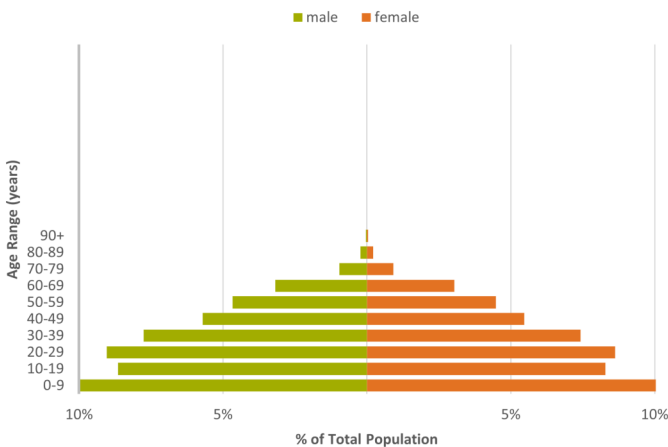


Fig. 2.1: Population distribution by age and gender (2023) [2.11]

The failure to provide adequate and affordable housing has fueled this informal expansion, leading to unregulated construction, unauthorized land use, and overcrowding in many districts. Many residents are forced to build multi-story informal housing units without structural safety considerations, increasing the risk of building collapses and infrastructure failures in these areas [2.1].

Coastal Vulnerability and Environmental Risks

Alexandria's location along the Mediterranean makes it highly vulnerable to the impacts of climate change, particularly rising sea levels, coastal erosion, and extreme weather events [2.2]. Already, parts of the city's coastline, including the Corniche, Montazah, and Stanley Bay, are experiencing regular flooding during winter storms. In the coming decades, scientific projections indicate that sea levels will rise by at least 50 cm by 2050, which could result in the permanent submersion of low-lying areas such as El Max, El Anfoushi, and parts of Lake Mariout [2.3].

Beyond flooding risks, saltwater intrusion is becoming a major problem, particularly in groundwater supplies and agricultural areas near Alexandria [2.7]. As seawater seeps into freshwater resources, it reduces agricultural productivity, contaminates drinking water supplies, and threatens biodiversity.

Another environmental concern is the pollution of Lake Mariout, which was once a major freshwater ecosystem supporting fisheries and local livelihoods [2.4]. Over the past few decades, the lake has become heavily polluted due to industrial waste, untreated sewage discharge, and urban encroachment [2.9]. Without urgent environmental remediation, the lake will continue to degrade, increasing Alexandria's vulnerability to climate-induced disasters [2.10].

Land Use and Zoning Issues

One of the most significant urban challenges in Alexandria is the poor distribution of land use and ineffective zoning regulations [2.1]. Historically, the city was designed around a compact urban core, with its development largely influenced by European colonial planning during the 19th and early 20th centuries. However, in recent decades, unregulated urban expansion has led to the emergence of densely packed districts, conflicting land uses, and severe overcrowding in many parts of the city [2.2].

There is a noticeable conflict between heritage preservation and commercial expansion in the city's older quarters, such as Mansheya and Raml Station. Many historic buildings have been either demolished or repurposed into high-density commercial zones, gradually losing Alexandria's architectural identity. Meanwhile, in the newer districts, such as Smouha and Montazah, urban planning is more structured. Still, these areas remain inaccessible to a large portion of the population due to high property prices and a lack of affordable housing policies [2.6].

A critical issue in Alexandria's urban planning is the encroachment of industrial activities into residential zones. In areas such as El Mex and El Dekheila, industrial complexes are located within close proximity to residential neighbourhoods, exposing thousands of residents to airborne pollutants, chemical waste, and water contamination [2.8]. The lack of clear zoning separation between industrial, residential, and commercial sectors has resulted in environmental hazards that disproportionately affect low-income communities near factories and ports.

Additionally, privatization of the coastline has further complicated land use issues. Many previously public spaces in many

areas have been converted into luxury resorts and commercial developments, limiting most residents' access to the Mediterranean shoreline [2.5]. Without stricter zoning laws and public land protection policies, Alexandria risks further diminishing its already limited public spaces [2.10].



Fig. 2.3: Residents of the informal settlements [2.16]

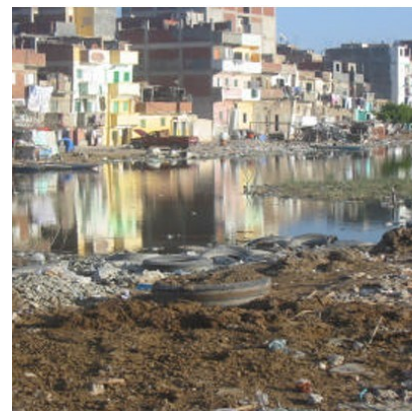


Fig. 2.4: Low standard of living and environment pollution in squatter settlements and Lake Mariout [2.3]

Urban Planning

Syed Monjur E Moul

Overall goal — vision	Specific goal	Action
Integrate and Improve Informal Settlements.	Ensure reliable utilities and secure land tenure.	Upgrade water, sewage, electricity; enforce building rules; legalize home ownership.
Protect the Coastline from Rising Seas.	Mitigate flooding risks and coastal erosion.	Extend seawalls/breakers; set development setbacks; install flood monitoring systems.
Increase Urban Green Spaces .	Meet World Health Organization greenery standards .	Encourage tree planting; create neighborhood parks; expand urban forests.
Strengthen Climate Resilience via Renewables.	Boost desalination and cut reliance on fossil fuels.	Increase desalination capacity; adopt solar/wind power; form an environmental authority.

Reforming Housing and Upgrading Informal Settlements

To tackle Alexandria’s housing crisis, solutions must address the uncontrolled high-rise construction in central areas and the rapid expansion of informal settlements on the city’s outskirts [2.2]. Currently, 1.36 million residents—40% of the city’s population—live in informal settlements [2.3]. The removal of strict building height limits in 2008 led to a surge in unregulated high-rise developments, overwhelming infrastructure and causing frequent power outages, water shortages, and sanitation problems [2.6]. The application of Law 119 of 2008 set height limits at 1.5 times the street width, with a maximum of 42 me-

ters, but enforcement has been weak [2.4]. To prevent further strain, zoning laws should be reinstated to control building heights in overcrowded districts, directing new high-density projects to well-planned areas with adequate transportation and public services [2.1].

While zoning reforms can enhance formal urban growth, informal settlements present a different challenge [2.9]. These settlements lack essential services such as clean water, electricity, and sewage systems, and 10 have no wastewater networks [2.5]. The urban upgrading initiative, covering seven selected informal settlements, aims to deliver 40 km of wastewater networks, 52 km of surface water drains, and 50 km of paved roads with curbs and sidewalks. Additionally, three multi-purpose community centers and a post office have been constructed to enhance local services.[2.5]

Instead of demolishing these neighborhoods, the city should focus on integrating them into the formal urban fabric through infrastructure improvements and legalizing land ownership [2.8]. A pilot slum-upgrading project in Favela, São Paulo, Brazil (Fig2.5), demonstrated that investing in informal settlements yields better results than displacing residents. Alexandria can follow a simi-

lar approach by prioritizing sewage systems, water pipelines, and road paving. Legalizing property ownership would encourage self-investment in housing improvements while providing residents with financial stability. Additionally, linking these neighborhoods to the city’s public transport system would reduce isolation and expand economic opportunities [2.10].

Egypt’s past experiences with slum redevelopment, such as the transformation of Manshiat Nasser in Cairo, demonstrate that comprehensive upgrading—including infrastructure investments, land reforms, and community participation—leads to higher property values and notable public health improvements [2.7]. A similar model in Alexandria could stabilize informal areas and convert them into functional, legally recognized neighborhoods rather than neglected, marginalized zones [2.3].

Improvements for Coastal Vulnerability and Environmental Risks

Alexandria should expand its wave breaker and seawall infrastructure to enhance coastal resilience beyond 5 km along the Corniche, Montazah, and Stanley Bay [2.6]. A 10 km extension is necessary to protect against the anticipated 50 cm sea level rise by 2050. A



Fig. 2.5: slum-upgrading project in Favela, São Paulo, Brazil.[2.17]

mandatory 500-meter construction setback should be enforced in high-risk coastal areas to prevent further urban encroachment into flood-prone zones. Implementing artificial sand dunes and wetland buffer zones would further protect the shoreline from erosion while enhancing biodiversity [2.8].

Alexandria must invest in a 60 km expansion of stormwater drainage networks to mitigate seasonal flooding, prioritizing areas such as El Max and Anfoushi, where flooding disrupts daily life. Permeable pavements and underground stormwater storage tanks should be incorporated into urban infrastructure to improve flood absorption capacity. A real-time flood monitoring system using tide gauges and automated flood barriers should be installed to enhance emergency response capabilities [2.9]. Alexandria should increase its desalination capacity to at least 500,000 cubic meters per day to combat saltwater intrusion in groundwater reserves, reducing reliance on over-extracted freshwater sources [2.4]. Additionally, a com-

prehensive aquifer recharge initiative should be implemented to reinject treated wastewater and harvested rainwater into key groundwater reservoirs, reversing salinity levels. Salinity monitoring stations must also be deployed in high-risk areas to track water quality deterioration and regulate excessive groundwater extraction [2.2].

Lake Mariout, which currently receives 500,000 cubic meters per day of partially treated effluent, needs immediate environmental rehabilitation [2.7]. The Eastern and Western Wastewater Treatment Plants should be expanded to provide an additional 600,000 cubic meters per day of secondary treatment, ensuring that only fully treated water enters the lake. Aeration systems and floating wetland biofilters should be deployed to restore oxygen levels and improve water quality. Furthermore, a 30-meter ecological buffer zone should be established around Lake Mariout, planted with pollutant-absorbing vegetation to reduce contamination [2.5].

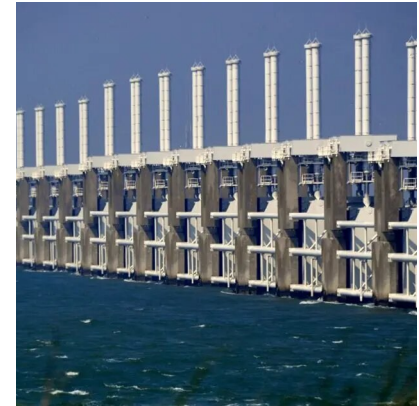


Fig. 2.7: Example sea wall of the Netherlands. [2.18]

The city must integrate renewable energy sources into its water management infrastructure to strengthen Alexandria's climate resilience. Solar and wind power should be used to operate desalination and wastewater treatment facilities, reducing fossil fuel dependency. Moreover, Alexandria must expand its urban green spaces from 684 to at least 1,500 acres, aligning with World Health Organization's 9 m² per capita recommendations [2.1]. Large-scale urban forestry and wetland restoration projects should be prioritized to enhance flood absorption capacity and improve air quality [2.6].

To ensure long-term environmental sustainability, an independent environmental monitoring authority should be established to track sea level rise, groundwater salinity, pollution levels, and flood risks, ensuring that intervention strategies are continuously refined [2.9]. Strict wastewater discharge regulations must be enforced alongside public education campaigns to promote water conservation and sustainable urban practices. Without these urgent interventions, Alexandria will continue to face escalating environmental degradation, increased flood risks, and irreversible ecosystem decline [2.8].



Fig. 2.6: Bishan, Singapore [2.19]

Green Infrastructure

Nikhila Sabu

Overview (2018)

Total green area	165.7 km² (2019) [3.1]
Green area ratio	9.4%
Green area per capita	0.98m² per person
Accessibility to green area	8% of population has access to green area within 300m linear distance [3.5]
Green infrastructure for stormwater/ flooding drainage	0.94% [3.6]
Existence of urban farming	Yes [3.5]

Enhancing Green Corridors

Green infrastructure denotes a system of natural and semi-natural networks that deliver ecological, economic, and social advantages by incorporating nature into urban spaces [3.2]. In contrast to conventional gray infrastructure that depends on engineered approaches, green infrastructure utilizes plants, soil, and water management techniques to improve sustainability and resilience. Instances consist of green roofs, urban woodlands, wetlands, and permeable

surfaces, which assist in temperature regulation, stormwater management, and enhancing air quality. With cities encountering issues such as climate change, pollution, and swift urban growth, green infrastructure is crucial in fostering environmental equilibrium and improving community living standards [3.6].

Green area shortages in Alexandria

The total area of Alexandria is 2679

Green, blue and grey infrastructure (2018)

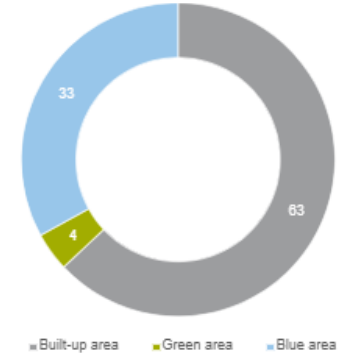


Fig. 3.2: Share of built up areas, green and blue infrastructure in %

km², of which about 163 km² are made up of agriculture [3.1]. Additionally, there are about 684 acres of gardens and green space in the city .The city's total area and population, which is around 0.9 m², make this green space (684 acres) seem inconsequential [3.3]. In addition to Alexandria's lack of green space, the increase in building violations after the 2011 Egyptian Revolution also led to a decrease in the city's total amount of green space. Rethinking how to benefit from the natural resources and corridors that are available to create the Green Network, which directs the city's expansion towards appropriate urban development, must go hand in hand with strict rules to prevent the loss of the precious green areas .[3.2]

Current state of green Infrastructure

The city of Alexandria is home to several urban parks, such as Al Montazah, Al Shalat, and Antoniadis parks. However, a lack of coordination in physical planning has led to a deterioration in the environmental quality and a decline in the availability of sustainable urban parks[3.5]. Additionally, the city's rapidly



Fig. 3.1: Aerial view of Alexandria (2025) [3.8]

growing population has expanded Alexandria, causing a gradual disconnect between people and nature. As a result, the residents have lost the opportunity to engage with natural spaces.

The City has adopted various green infrastructure practices, both within the Combined Sewer Service (CSS) area and across the entire city. These practices were selected as a key strategy to reduce stormwater runoff and remove pollutants. To date, green infrastructure techniques are managing runoff from about 94 acres of stormwater, with roughly 1.75 acres located within the combined sewer service area.

Access to green spaces

Access to green spaces in Alexandria is notably limited. Only 8% of Alexandria's residents live within 300 meters of a green space. This is well below the WHO's suggested minimum of 9 m² of green spaces per person [3.3]. Efforts

are underway to enhance Alexandria's green spaces. Initiatives like the revitalization of Al-Shalat Park (Figure 3.3) focus on improving park usability and accessibility for the local community. Additionally, projects such as 'Redefining public parks' aim to rejuvenate public gardens, promoting their use and conservation.

Urban Farming

Urban agriculture in Alexandria can significantly contribute to improving the city's green infrastructure by combining food production with strategies for ecological and climate resilience. Green roofs, vertical farms, and community gardens can act as versatile green areas, enhancing stormwater control, lowering urban temperatures, and boosting biodiversity. These areas can serve as carbon sinks, enhancing air quality and providing social and economic advantages to nearby communities[3.5]. Integrating urban agriculture into Alexandria's green infrastructure strategy will

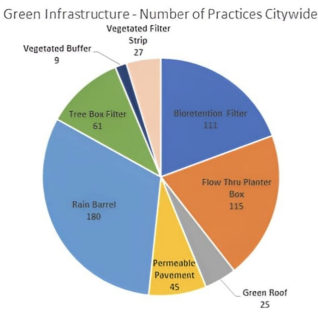


Fig. 3.4: Green infrastructure practices adopted in Alexandria [3.6]

enable the city to develop a more sustainable urban setting that harmonizes food production with ecological recovery and climate resilience.

Environmental Issues

Nestled between the sea, lagoons, and ancient lakes, the city is situated on a T-shaped peninsula.[3.4]. Because of the city's high elevation below sea level, drainage and floods are major problems. Due to environmental risks and physical constraints, future urban growth must take place in non-low-lying areas. The informal districts of Alexandria, which are home to one-third of the city's population, have also seen a sharp increase in the number of unplanned constructions and illegal settlements after the 2011 Egyptian revolution. Both people and the city's infrastructure are considered to be seriously endangered by these violations.[3.2]. Although Alexandria's air quality is typically acceptable when compared to air quality standards, this does not imply that air pollution does not exist in other areas. Alexandrians now have to deal with air pollution brought on by clogged roadways and unpermitted structures.

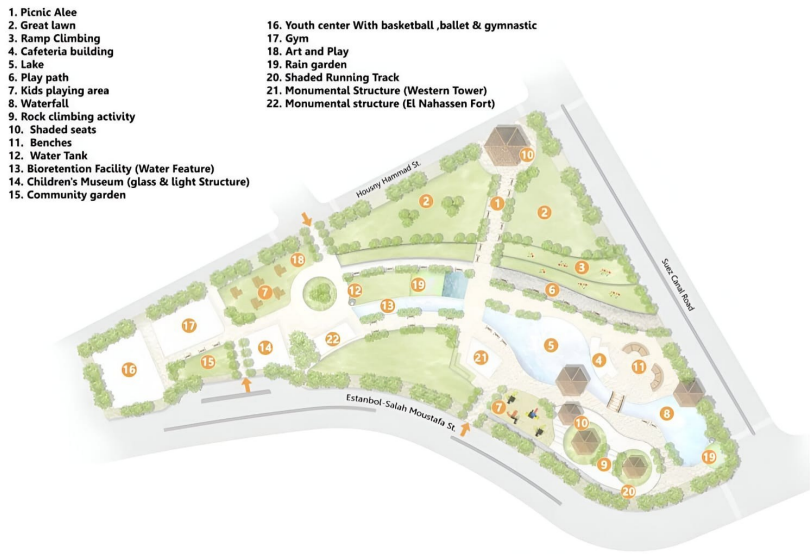


Fig. 3.3: Proposed plan for revitalization of Al-Shalat Park [3.9]

Green Infrastructure

Nikhila Sabu

Overall goal — vision	Specific goal	Action
From green belt to green network	Increase the green area by 9 m ² per person by 2040	Increase the proportion of public parks and open spaces
Transform Alexandria into a sponge city	Stormwater management and reduce drainage issues to 45 % by 2030	Implement tree box filters, permeable pavements and rain gardens to reduce flood and drainage issues.
Enhancing waterfront and green corridors	Protect and rehabilitate wetlands and dune systems	Establish node system along the Mahmoudeya canal to provide a good ecological network.
Create healthier, more livable spaces	Reduce carbon footprint to 40% by 2035	Expand public transportation, electric buses and non-motorized mobility

From Greenbelt to Green Network

In the southeast, the city of Alexandria is surrounded by greenbelts of farmed land that may be developed in a manner similar to Ebenezer Howard's idea. The purpose of the greenbelt is to stop more urban development and to stop the degradation of nearby agricultural area. The Alexandria Green Network may be established using the idea of a city greenbelt, as demonstrated by the Ottawa, Canada, Greenways network. [3.2]. Ottawa's greenbelt was designed to include a range of land uses inside its borders, including farms, leisure spaces, natural areas, and property put aside for



Fig. 3.5: Waterfront walkway along Lake Ontario [3.10]

future purposes. In order to connect the greenbelt to other urban and rural transit routes, a network of paths and picturesque drives was suggested.

Enhancing waterfront and green corridors

The Mahmoudeya Canal is a vital natural corridor that runs through the urban fabric of Alexandria's eastern region. Through his attempts to modernize Egypt overall in the 19th century, Muhammad Ali built the Mahmoudeya Canal to supply fresh water to the city and to connect the city to the Nile River by providing direct access to and from the Nile at Cairo [3.2].

By using the idea of a city canal, the Alexandria Green Network might be achieved, as demonstrated by the Meewasin Valley, Canada, greenways network. Throughout the valley, eight nodes were to be developed in order to implement the Meewasin greenways' conceptual principles [3.2]. The metropolitan area's nodes were chosen to support the growth of commercial, cultural, and business establishments in addition to leisure activities. It is feasible to walk, hike, or ride between all of these nodes because of the greenway

system that was put in place between them.

Alexandria is well-known for its waterfronts, a strip of land that is bordered to the north by the Mediterranean Sea and to the west and south by Lake Maryout. Similar to Toronto, Canada's greenway development concept, the idea of these waterfronts might be expanded to create the Alexandria Green Network. In order to connect each neighborhood, arterial greenways are built as part of the proposed two-tiered networks. A waterfront walkway along Lake Ontario's coast (Figure 3.5) is part of one of these arterial networks, while another connects local connectors in each municipality and spans the whole area. The development of the local connections must take into account the goals of the municipalities and the prospective land availability. In order to connect community parks and tributary valleys and create several recreational trail loops, rail lines and other utility routes aid in the improvement of these connections.

Measures to reduce Carbonfootprint

Several cities around the world have successfully implemented measures to

reduce their carbon footprint, offering valuable lessons for Alexandria. For example, Copenhagen has invested heavily in green transportation, with over 50% of its residents commuting by bike [3.4]. Implementing a similar strategy in Alexandria by expanding bike lanes and pedestrian-friendly streets could significantly cut emissions from personal vehicles. Additionally, enhancing public transportation through electric buses and tram systems, as seen in cities like Bogotá, would reduce reliance on fossil fuels while improving air quality.

Another effective strategy comes from Singapore, which has integrated green infrastructure into urban planning. The city has mandated green roofs (Figure 3.7), vertical gardens, and extensive tree-planting initiatives, helping absorb carbon dioxide and cool the urban environment [3.3]. Alexandria could adopt similar policies by promoting green building designs, increasing urban forests, and utilizing nature-based solutions for

coastal protection. Furthermore, cities like San Francisco have successfully minimized landfill waste through strict recycling policies and composting programs, which could be adapted to Alexandria's waste management system to reduce methane emissions[3.7].

Stormwater Management

Modifying green and gray infrastructure approaches to fit Alexandria's urban setting and climate is essential for effective stormwater management [3.6]. Surface runoff can be diminished and groundwater recharge improved by employing methods such as rain gardens, bioswales, and permeable pavements. Green roofs and rainwater collection can alleviate flooding and promote water conservation in Alexandria, where development hinders natural drainage, enhancing biodiversity and absorbing rainfall by developing urban wetlands and expanding green areas. An alternative method to improve storm-



Fig. 3.7: Green roof practices in Singapore [3.11]

water management is by upgrading current infrastructure with flood-resistant designs, such as raised roadways and permeable pathways.

Another essential strategy is the restoration of natural water bodies and wetlands. Wetlands function as natural sponges, cleansing pollutants and lowering flood risks through the absorption of surplus water. For instance, restoring the wetland ecosystem of Lake Maryout could boost stormwater retention and increase biodiversity.

Conclusion

For effective execution, green infrastructure should be integrated into urban planning policies, enhanced by collaboration across disciplines, and supported by funding from both public and private sectors. Community involvement and awareness initiatives will be essential for achieving lasting success and sustainability. Improving Alexandria's green infrastructure is both an ecological imperative and a strategic method for building a sustainable city.

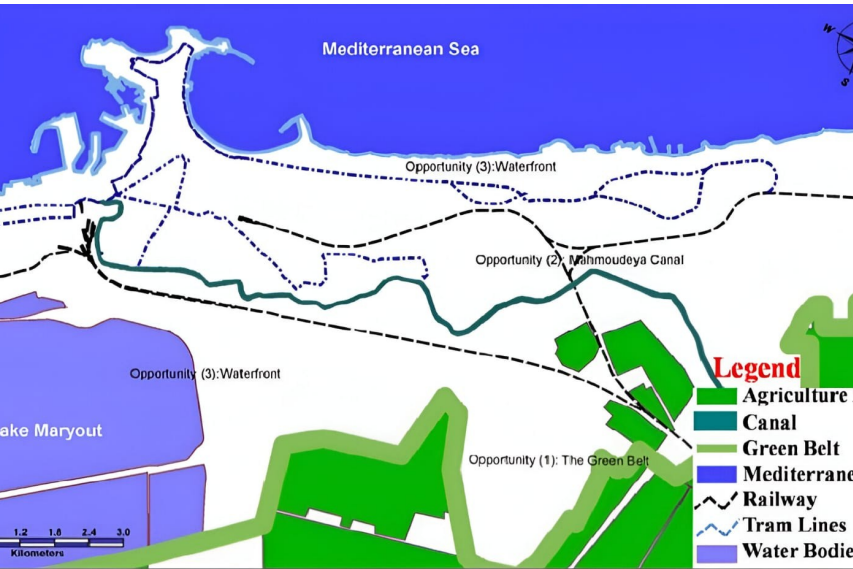


Fig. 3.6: Plans for Alexandria to have a Green Network [3.2]

Mobility

Riyamol Roy

Overview (2019)

Public transport use rate	26 % [4.1]
City area covered by public transport	~ 60 % [4.5]
Walking rate	12 % [4.2],[4.1]
Average commuting distance (walking)	4.12 km/day [4.12]
Cycling rate	8 % [4.1]
Cycling infrastructure	~ 2050 km of cycling lanes[4.12]
Density of private vehicles	605 motor vehicles per 1.000 people [4.12]
Traffic index	199.44 [4.12]

Overview

Mobility is an important aspect of modern life, shaping how people move within and between urban and rural areas. Efficient mobility systems are essential for economic growth, social connectivity, and environmental sustainability in a city.[4.1]

Alexandria is the second metropolitan area in Egypt with a coastal strip shape. Alexandria's ancient infrastructure, coastal geography, and expanding urban needs all influence mobility. The Corniche is a picturesque coastal route in the city that runs along the Mediterranean



Fig. 4.1: Corniche waterfront road –major corridor of transport [4.8]

sea. It is the major corridor for transport in Alexandria and one of the busiest routes in the city as it offers a variety of transportation options. Also serving as a major link between Alexandria's eastern and western regions, the Corniche facilitates mobility for tourists as well as local commuters.[4.3]

Modal Split

The rising demand for transportation in Alexandria is a direct result of rapid urbanization, economic expansion, and high population density, particularly in central areas. However, the city's road infrastructure and public transport services have not been adequately developed to keep up with these growing needs. Alexandria's overall road network length did not develop at a significant rate over the previous 20 years. It increased by an average of only 0.69% per year.[4.1]

The existing public transport system is plagued by poor planning, unreliable schedules, and insufficient capacity, leading to a steady decline in its appeal, despite being the primary mode of transport in the 1980s. In Alexandria , only 18% of motorized daily trips were

Modal split of transportation (2019)



Fig. 4.2: Means of Transportation in Alexandria in % [4.1]

completed using public transport in 2016, compared to 54% in 1985.[4.5] The primary reason is the deterioration of public transport service quality. The funding from government was mainly for the construction and maintenance of roads and so the public transport did not receive any benefits. Additionally, public transport was subjected to many unreasonable decisions. For instance, few public transport were cancelled on assumption that it can increase the delays for private vehicles.[4.1] Presently, the road system has become inadequate for all; traffic, parking and pedestrians.

All these factors have significantly altered the transport modal split over the past decades, leading to a sharp increase in the use of private vehicles and informal public transport.

Public transport

Public transportation in Alexandria, Egypt, is a vital yet struggling component of the city's mobility network. Only approximately 30% rely on public transport at present[4.1]. The city offers a variety of transportation options like tram system, buses, railway services as well as ferries. Alexandria's tram network is one of the oldest in the world

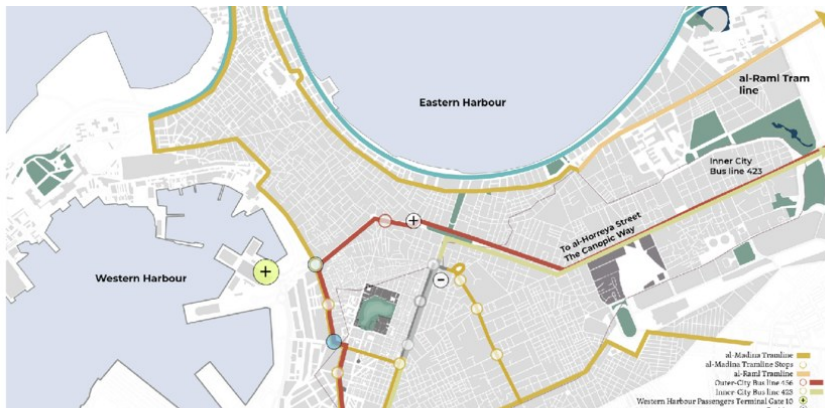


Fig. 4.3: Public transportation network in Alexandria [4.5]

which comprises of two main lines: Al Raml and Al Madina. The city has bus services that are affordable and is also well connected to other major Egyptian cities via national railway network. Additionally, ferries provide scenic and practical mode of transport, mainly operating along Eastern Harbor and connecting various coastal points.[4.2] Overcrowding is a major issue during peak hours, making travel uncomfortable for passengers. Inconsistent schedules and lack of real-time tracking make difficulties for passengers to plan their trips efficiently. Apart from that, the accessibility element [4.5] (distance between destination and the transportation station) also plays a crucial role. Due to these reasons majority of people in the city rely on private vehicles.

Private vehicles

In Alexandria, private vehicles are one of the most used modes of transportation (nearly 60%) due to their convenience and flexibility[4.2]. Many residents prefer cars and motorcycles over public transport to avoid delays, overcrowding, and inconsistent schedules. The passenger car ownership has an average yearly growth factor equals 1.31%[4.1]. However,

the high number of private vehicles contributes to traffic congestion, especially in busy areas like the Corniche and city centre. Parking shortages and rising fuel costs are ongoing challenges for private vehicle owners.[4.12]

Public transport in Alexandria has struggled to keep pace with the rapidly growing demand for mobility. Since the 1980s, Egyptian law has permitted private vehicles with up to 14 seats to operate as collective taxis in urban areas, with routes and fares regulated by local authorities. The collective taxi network spans the entire city, expanding particularly in areas where bus services are insufficient [4.1],[4.2].

Active mode

Active modes of transportation in Alexandria, such as walking and cycling, are less common but still play a role in daily commuting. Many residents and tourists walk along the Corniche. However, cycling is not commonly adopted due to a lack of dedicated bike lanes, heavy traffic, and safety concerns. Pedestrian infrastructure in some areas is underdeveloped, with uneven sidewalks and limited crossings, making walking less

convenient in certain parts of the city.

Challenges

Alexandria experiences significant traffic jams, particularly in crowded spots such as the Corniche, because of a rising amount of private cars and restricted road infrastructure. The public transportation network faces inefficiency, as old buses and trams combined with irregular schedules render commuting unpredictable. Furthermore, active mobility is insufficiently developed, as there is a scarcity of infrastructure that is friendly to pedestrians and cycling paths, rendering walking and biking unsafe.



Fig. 4.4: Historic tram system in Alexandria [4.9]

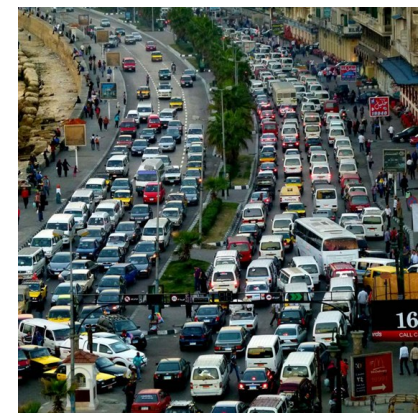


Fig. 4.5: Car dependency issue in the city [4.10]

Mobility

Riyamol Roy

Overall goal — vision	Specific goal	Action
Reduce traffic congestion	Increase public transport ridership by 40% by 2030	Implementing Bus Rapid Transit (BRT) System ,discounted fares
Smart and efficient traffic management	Reduce travel delays by 25% by 2030	Real-time monitoring using sensors, automatic traffic signals.
Green transport system	Reduce carbon emissions by 50% by 2032	Electric vehicles, expand active transport, emission regulations

Increase public transport usage

To address traffic issues more effectively, there should be a shift away from the reliance on personal vehicles toward investing in a quality public transportation system [4.2]. The target to be accomplished within this framework is an increase in public transport usage up to 40% by the year 2030. [4.13] A Bus Rapid Transit (BRT) System is one of the best options because it has the means to immediately achieve these goals, as it provides reliable and fast service through the use of special bus lanes. Alexandria could replicate the model used in Bogotá, which efficiently expanded its public transportation system with the Tri-Milenio BRT system and simultaneously reduced traffic con-

gestion. Trans-Milenio is a BRT system in Bogotá that utilizes high capacity buses and operates in dedicated lanes, allowing for efficient bus transport between regions while collaborating with other urban transport systems. An Alexandria BRT system, with overly dedicated BRT lanes on the Corniche and Abu Qir Street would not only make public transport more efficient and trustworthy, but also lessen the reliance on private vehicles. [4.12] With a similar system, public transport would become more efficient and aid in reducing congestion in Alexandria.

The price of using public transport is another barrier that needs to be solved. The government needs to make public transport integrated with the rest of the city to encourage its usage – like the abolishing of fares partly or providing discount tickets for different public transport vehicles. Increased accessibility of public transportation has the potential to attract students, low-income individuals, and employees who depend on these services.[4.8]

Smart and efficient traffic management

Due to a lack of attention towards effective traffic management in Alexandria, there is a considerable amount of time wasted during transit, alongside excessive fuel consumption and pollution. Currently, several traffic intersections in

Alexandria have poorly managed traffic lights that do not operate well with the existing uncontrolled congestion nodes, causing frequent issues in movement. In light of this, city infrastructure may benefit from Smart & Efficient Traffic Management [4.4],[4.7]. One of the key goals for this initiative is to reduce traveling delays by 25% by the year 2030. One such model that has shown success in adopting this transformation is Singapore's Smart traffic Management System that employs real-time sensors, AI-based congestion prediction, and adaptive traffic lights to ensure a steady flow of traffic. Alexandria can implement a similar system by Intelligent traffic sensors installation, signal coordination enforcement, and congestion anticipation and remediation through data analytics[4.7].

Moreover, commuters could get live traffic updates and suggested alterations to their routes through digital signboards and mobile applications allowing them to optimize their trips. Additionally, giving buses and emergency vehicles priority passes at traffic signals could promote greater use of public transportation which in turn will reduce traffic congestion[4.13]. By adopting these measures, Alexandria can build a smarter, more efficient, and sustainable city.

A green, low-emission transportation system is essential for minimizing car-

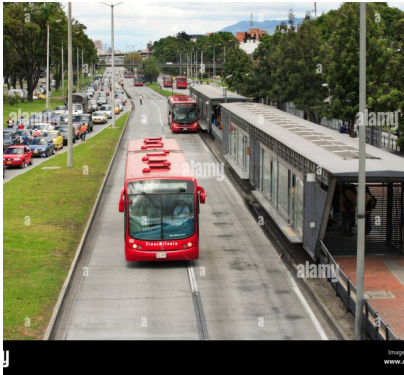


Fig. 4.6: Trans-Milenio Bus Rapid Transport System [4.12]

bon footprints and promoting sustainable urban mobility. Oslo, Norway, serves as a prime illustration of a city committed to reducing carbon emissions via a comprehensive approach that encompasses extensive electric vehicle (EV) usage, major growth of active transportation systems, and the implementation of strict emission regulations. Considering that transportation is a significant source of urban emissions, Oslo's strategy emphasizes decreasing reliance on fossil fuels while putting resources into sustainable infrastructure. This shift not only lessens climate change but also improves air quality and decreases congestion. An essential element of Oslo's strategy is the encouragement of active transportation, including biking and walking, in conjunction with funding for public transit [4.7]. The city has greatly enhanced its cycling infrastructure, converted important zones for pedestrians, and incorporated electric buses and trams within its public transit system. More than 60% of daily journeys in Oslo are currently conducted using eco-friendly transporta-

tion methods. Likewise, Alexandria can promote active transit by creating specific cycling paths, enhancing pedestrian facilities, and upgrading its public transit system with electric buses and trams. Limiting vehicle access in highly crowded zones, like downtown and the Corniche, might enhance sustainable transport and decrease emissions [4.6].

In addition to this, Oslo has enacted stringent EV policies and rigorous emission controls. Alexandria can adopt comparable strategies by encouraging EV purchases, creating charging infrastructure, and establishing low-emission zones in high-density regions [4.7].

Conclusion

The increasing need for transportation in Alexandria, influenced by urban growth and economic development, has surpassed the progress of road infrastructure and public transit. The city encounters major transportation



Fig. 4.8: : Smart traffic management in Singapore (conceptual simulation) [4.11]

issues, such as traffic jams, insufficient public transportation, and a strong reliance on private vehicles. To tackle these challenges, Alexandria can adopt thorough measures to improve public transportation effectively. Creating a Bus Rapid Transit (BRT) system offers an efficient substitute for private cars, alleviating congestion and shortening travel durations , while implementing low-cost fare strategies can motivate more community members to opt for public transport. Intelligent traffic management systems can improve traffic flow, minimize delays, and boost urban mobility by employing real-time data and adaptive signaling. Enhancing active transport infrastructure, such as bike lanes and pedestrian-friendly areas, will encourage sustainable and environmentally friendly mobility choices. Moreover, shifting to an eco-friendly transport framework through the adoption of electric vehicles and the creation of low-emission areas can aid in lowering carbon emissions and enhancing air quality. By implementing effective and sustainable mobility options, Alexandria can develop a cleaner and more reachable urban transport system.

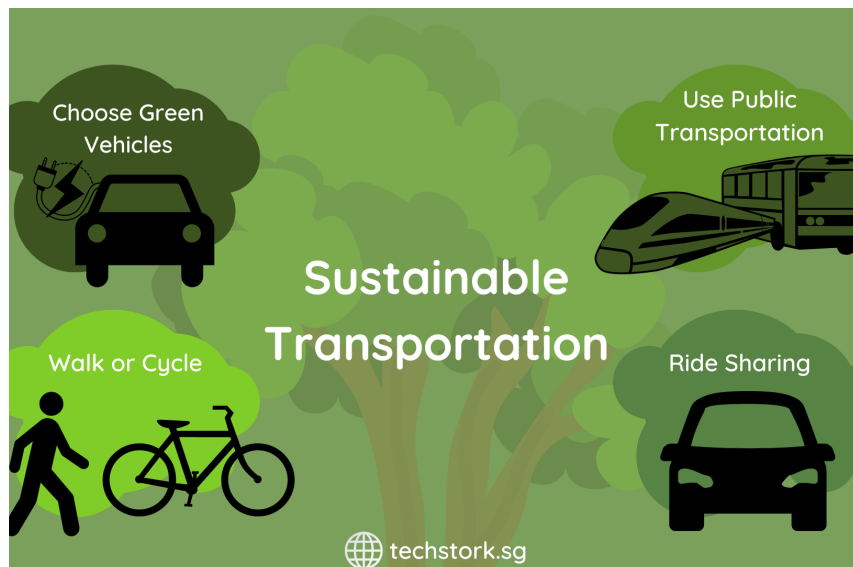


Fig. 4.7: Green transport solutions, Oslo ,Norway [4.10]

Energy

Arya Aji

Overview (2022)

Primary energy demand	102 million TOE [5.1]
Energy consumption per capita	0.9 TOE per capita [5.1]
Energy consumption per unit of GDP	180 TOE/\$1M [5.1]
Renewable energy share	15 % [5.2]
Access to electricity	99.98 % of the population [5.3]
Gross electricity consumption	215.8 TWh [5.4]
Electricity consumption per capita	1.6 MWh [5.4]
Total CO2 emissions	13 Million of tCO ₂ [5.4]
CO2 emissions per capita	2.6 tCO ₂ per capita [5.4]
CO2 emissions per unit of GDP	470 tCO ₂ /\$1M [5.4]

Energy of Alexandria

The natural system know as the “greenhouse effect/” regulates the temperature on earth. Human activites have the potential to disrupt the balance of this system. As human societies adopt increasingly sophisticated and mechanized lifestyles, the amount of heat trapping gases in the atmosphere have been increased. By increasing the amount of these gases, humankind has enhcnaced the warming capability of the natural greenhouse effect. It is the human-induced greenhouse effect that cause environmental concern. It has the potential to warm the planet at a rate that has never been experienced in human history. This intensifies the natural greenhouse effect causing Global Warming or Climate Change. Global Warming of just a few degree would present an enormous change in climate. [5.5]

Alexandria, Egypt’s second-largest city is a vital contributor to the nation’s economy and energy landscape. As of 2021’ Egypt’s electricity generation heavily relies on natural gas accounting for approximately 81% of the total, with

oil contributing about 7.5%, hydroelectric power 6%, wind 3% and solar 2%. [5.6]

In Alexandria, the energy infrastructure includes significant natural gas, power plant such as te El Nubaria Power Plant with a capacity of 2,250 MW and the Sidi Krir Power Plant at 750 MW.

Alexandria’s energy and environmental environment encounters major obstacles, especially in the transportation sector and aquatic ecosystems. The city’s elevated energy intensity arises from its significant dependence on road transportation, characterized by inefficient, hydrocarbon-powered vehicles resulting in excessive fuel use and carbon emissions. Enhancing public transit systems, updating vehicle fleets, and incorporating alternative fuels are vital measures for minimizing energy waste. [5.7]

At the same time, climate change increasingly endangers Alexandria’s coastal ecosystem, as rising sea temperatures drive fish species northward* and raise salinity levels in coastal lakes, threatening local fishing industries. To address these challenges, the city is funding eco-friendly transportation options, coastal protection initiatives, and

Primary energy supply sources (2021)

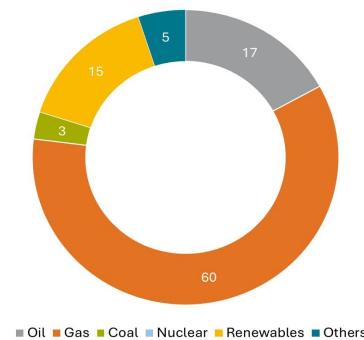


Fig. 5.1: Percentage of various energy source of Alexandria 2021 [5.4]

Electricity generation sources (2022)

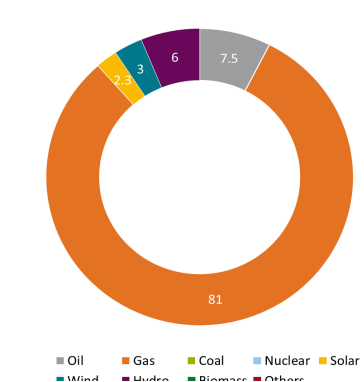


Fig. 5.2: Percentage of various electricity sources of Alexandria [5.4]

CO2 emissions per capita

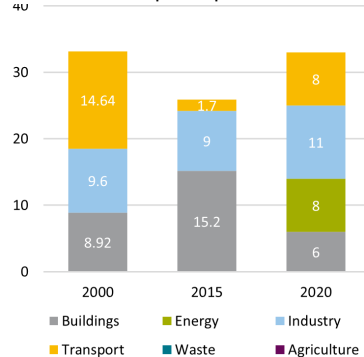


Fig. 5.3: CO2 emission of Alexandria [5.3]

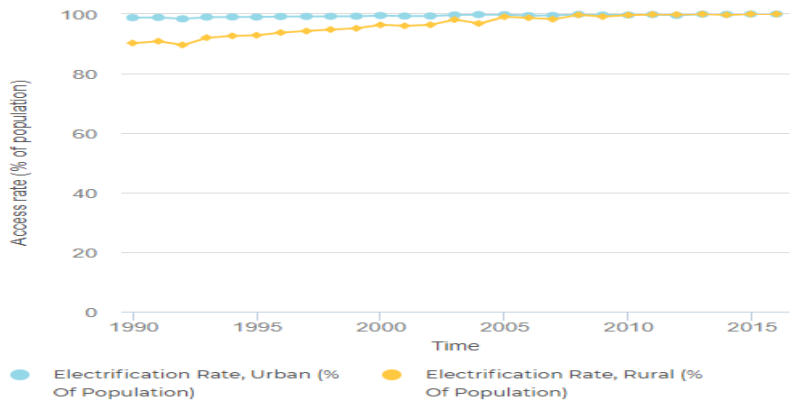


Fig. 5.4: Egypt Energy Access Urban and Rural [5.5]

innovative aquaculture methods, in line with its larger goals for energy conservation, ecological sustainability, and climate resilience. [5.6]

Emission trends

Alexandria, Egypt, is estimated to emit around 19.7 million tonnes of greenhouse gases (GHG) annually, with the main contributors being electricity usage (nearly 9.7 million tonnes CO₂e), natural gas (nearly 5.9 million tonnes CO₂e), gasoline (nearly 2.3 million tonnes CO₂e), and diesel (nearly 1.8 million tonnes CO₂e). [5.7]

In the absence of intervention, emissions are predicted to increase because of higher energy needs and industrial activities. In reply, Egypt's Nationally Determined Contribution (NDC) aims for a 33% reduction in emissions from the electricity sector and 65% in oil and gas by 2030 relative to the business-as-usual scenario. [5.8]

The National Climate Change Strategy 2050 seeks to attain 42% renewable energy in electricity production by 2030, as well as enhanced energy efficiency and sustainable urban development. [5.9] The significant industrial activity

and dependence on fossil fuels in Alexandria underscore the pressing need for regional execution of these national strategies to reduce emissions and advocate for a sustainable future. [5.9]

The energy use and emissions situation in Alexandria, Egypt, shows that the city encounters considerable difficulties regarding emission reduction. The energy-heavy sectors within the city, such as transportation, manufacturing, construction, and residential structures, greatly contribute to greenhouse gas emissions. [5.10]

The manufacturing and construction sectors are the biggest contributors to emissions, accounting for 42% of the total. In addition to this, residential structures contribute 26%, while the transportation sector represents 18% of the overall GHG emissions. Furthermore, waste management is responsible for 5%, whereas commercial and institutional structures represent 9% of emissions. Alexandria's dependence on fossil fuels, especially oil and natural gas, worsens its carbon emissions, highlighting the need for immediate action to shift towards renewable energy and sustainable urban planning. [5.12]

To reduce emission and enhance sustainability in the energy sector, Alexandria, Egypt is focusing on expanding renewable energy and improving energy efficiency. These initiatives support Alexandria's goal of building a more resilient and low-carbon energy system. [5.4]

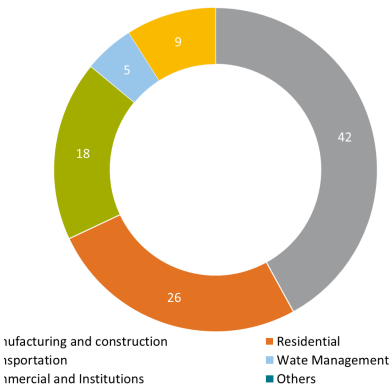


Fig. 5.5 Sector wise share of GHG emissions, 2021-22 [5.1] [5.6]

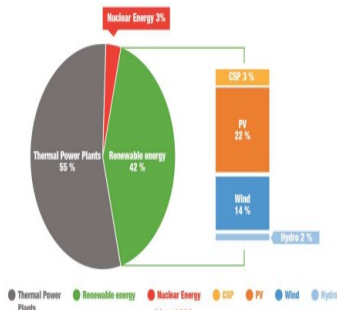


Fig. 5.6: Alexandria Target Energy Mix 2035 [5.6]

Energy

Arya Aji

Overall goal — vision	Specific goal	Action
Increasing Renewable Energy Penetration	Aims to generate 42% of its total electricity from renewables	Expanding solar and wind projects by using power grid
Enhance Energy Efficiency	Smart infrastructure and Energy Efficient Projects	1.Upgrading Public infrastructures 2. Encouraging Energy Building Standards
Development of Nuclear Power	Empower Energy Security	Construction of El Dabba Nuclear Power Plant

Vision for sustainable energy

In 2021, the overall amount of power generated by renewable energy resources is approximately 19.2 GW. Aiming for 50.5 GW in the years 2029/2030 and 62.6 GW in the years 2034/2035, the government needs to promote renewable energy production to a total of 62.6 GW by 2034/2035 [5.13] Renewable energy will account for 42% of total Egyptian electricity output in this scenario by 2035, according to the World Energy Outlook. [Figure 5.7] depicts the evolution of installed renewable energy capacity over time

The 2035 goal of 42% RE targets 14% from wind, 2% from hydropower, 22%

from Photo Voltaic (PV) 22%, 4% from Concentrating Solar Power (CSP), 3% from Nuclear Power Plants (NPP) and 55% from thermal energy from fossil fuel (coal, gas and oil) [5.6]

Additionally, Alexandria seeks to improve energy efficiency by incorporating smart infrastructure, energy-saving initiatives, and industrial optimization to minimize energy usage and environmental effects. This involves implementing advanced metering systems and smart grid technologies to enhance energy distribution, advocating for energy-efficient architectural designs, and motivating industries to embrace upgraded equipment to reduce carbon emissions. [5.10] These initiatives correspond with Egypt's National Energy Efficiency Action Plan (NEEAP) and wider sustainability objectives aimed at modernizing energy systems and enhancing resource utilization.

Another major vision in Alexandria is to enhance energy security through the incorporation of nuclear energy into Egypt's power system. This vision seeks to create a reliable, diverse, and low-carbon energy system that minimizes reliance on fossil fuels and promotes long-term sustainability. Through the use of nuclear energy, Alexandria aims to foster economic development, industrial growth, and a robust energy infrastructure that caters to rising demand

while maintaining environmental sustainability. [5.13]

Challenges in achieving the vision

The expansion of renewable energy to meet the 42% target by 2035 is hindered by high capital costs, underdeveloped infrastructure, and the continued reliance on natural gas due to recent discoveries. The integration of wind, solar, and hydropower also requires significant grid modernization and storage solutions to manage intermittent energy generation effectively. Additionally, policy and regulatory frameworks need further development to attract private sector investments and ensure a smooth transition to a diversified energy mix.[5.13]

According to Egypt's National Energy Efficiency Action Plan (NEEAP), improving energy efficiency in Alexandria is fraught with difficulties, including institutional restrictions, inadequate monitoring, and insufficient public recognition. [5.10] With a total capacity of 4.8 GW, Egypt's first nuclear power plant, the El Dabaa Nuclear Power Plant (NPP), will start operations in 2026. It is situated 140 km west of Alexandria. The plant, which was financed by a \$25 billion loan anticipated to help Egypt's long-term sustainability goals by operating for more than 60 years with low CO₂ emis-



Fig. 5.7: The evolution of Alexandria renewable energy capacity in GW [5.13]

sions. It may slow down the spread of renewable projects, which raises questions about how to achieve a diversified and balanced energy transition. [5.13]

Action towards achieving the vision

Egypt is making significant progress toward achieving its objective of producing 42% of its electricity from renewable sources by 2035. To achieve this, the nation is implementing a focused approach known as the Integrated Sustainable Energy Strategy (ISES) 2035, which aids in drawing investments and establishing appropriate policies for the advancement of clean energy.

A significant achievement to date is the Benban Solar Park, among the largest solar farms globally, generating 1.65 GW of electricity. Egypt is collaborating with global partners and investors to accelerate renewable initiatives and introduce new technologies. Simultaneously, the nation is upgrading its power grid to enable more efficient use of solar and

wind energy. These initiatives are crucial for decreasing dependence on fossil fuels, lowering emissions, and creating a cleaner, more sustainable future. Benben is currently world's 4th largest solar park [5.15]

Improving Alexandria's public infrastructure is vital for boosting energy efficiency and sustainability, which involves modernizing electricity networks to facilitate renewable energy integration, implementing smart grid technologies for improved energy management, and enhancing public transportation with more eco-friendly options. The implementation of advanced metering systems (AMS) will aid in tracking and enhancing electricity usage, minimizing energy wastage, while funds directed towards energy storage technologies will guarantee a reliable power supply, especially for variable resources such as solar and wind energy (NEEAP). Moreover, Alexandria is advocating for energy-efficient construction practices by implementing sustainable building

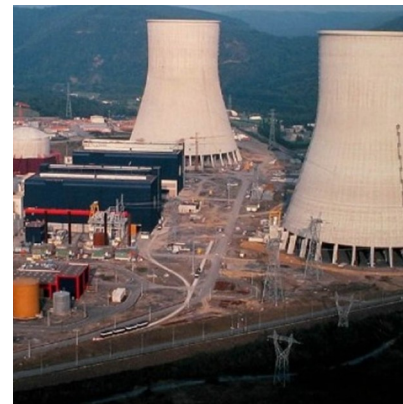


Fig. 5.9: El Dabba Nuclear Power Plant [5.13]

codes, requiring insulation, and promoting solar panel use, backed by tax breaks and grants to stimulate environmentally friendly designs. In the meantime, the El Dabaa Nuclear Power Plant (NPP), situated 140 km west of Alexandria, will deliver 4.8 GW of consistent, low-carbon energy. For successful outcomes, rigorous regulatory supervision, employee training, and adherence to international safety standards will be essential in harmonizing the growth of nuclear energy and renewable energy, decreasing reliance on fossil fuels, and reinforcing Egypt's long-term sustainability pledges. [5.12] [5.13]

Conclusion

Alexandria faces energy and environmental challenges due to its reliance on fossil fuels, inefficient transport, and high emissions. The city is working to expand renewable energy, improve efficiency, and modernize infrastructure to support sustainability. Egypt aims to reach 42% renewable energy by 2035, with smart grids and energy-efficient initiatives playing a key role. Investing in eco-friendly transport, coastal protection, and emissions reduction will help secure a cleaner and more resilient future.

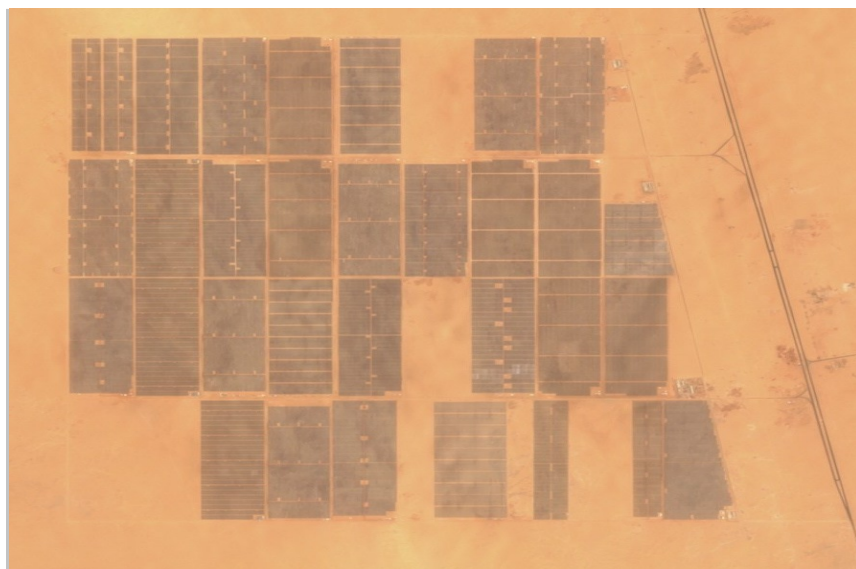


Fig. 5.8: Benban Solar Park [5.15]

Water

Moaaz Ramadan

Overview (2019) [6.5],[6.12],[6.13],[6.14]

Water Availability per Capita (1960s)	2,200 m ³ per capita per year.
Water Availability per Capita (2018)	570 m ³ per capita per year
Current Water Resources (2019)	610 m ³ per capita
Alexandria's Water Needs (2017) Public water supply	2.9×10^6 million m ³ /day
Alexandria's Water Needs (2017) Industrial water demand	427×10^6 m ³ .
Desalination Efficiency	80% reduction of freshwater production cost compared to conventional plants.

Stress on resources

Worldwide, water resources problems will be one of the most significant economic and social issues of the century. As there are limited water resources due to its fixed share of the Nile water, Egypt is one of the countries that will face great challenges, and limitation of groundwater, rainfall and desalination water. Climate change plays an important role in imposing an additional burden on water availability and accessibility in Egypt. The upper Nile basin countries' increased competition on water (The Grand Ethiopian

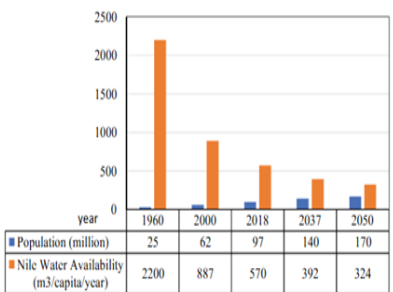


Fig. 6.1: Availability of water supply in Egypt [6.1]

Water supply sources (2019)

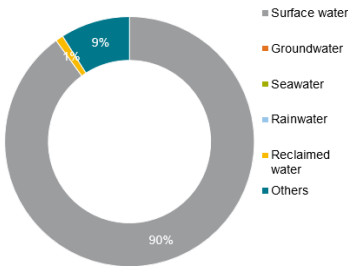


Fig. 6.2: Share of water supply sources(2019) [6.0]

Renaissance Dam) may lead to water shortage. Although, Lake Nasser budget of water represents the national freshwater bank of Egypt, depending on the water inflow through the Nile discharge and the water outflow through the High dam, there is still water loss by the evaporation and limited water seepage from Lake Nasser. The Tushka project is part of a national scheme to reform this pattern of skewed population potentials distribution [6.1].

Egypt is facing major challenges because of its limited water resources and rapidly growing population. Water use per capita has decreased from 2200 m³/capita/year in the 1960s to 570 m³/capita/year in 2018, and it is projected to be only 324 m³/capita/year in 2050 (Fig. 6.1). Thereby, Climate Change is projected to cause even greater concerns throughout the country, with severe environmental, social, and economic implications. The current water availability in Egypt is insufficient to address the required water demand for agriculture, industry, domestic use, and others, where transboundary water agreements commitments do exist. Moreover, the threats of reducing the share of the Nile in Egypt arise due to the development

plans in the upper Nile Basin countries. It is worth mention that CC can provide challenges and opportunities for the Nile River Basin (NRB) countries to collaborate for reducing the adverse impacts of Climate Change. [6.2].

Egypt, with available water resources of 610 m³ per capita as of 2019, is among the countries that fall below UN's water poverty line (estimated at 1,000 m³ per capita). Egypt also suffers from a large food-production deficiency, forcing it to import about 55 percent of its current food needs. This figure is expected to climb to 75 percent after the Grand Ethiopian Renaissance Dam (GERD) is complete because of the huge quantity of water held back during the filling of the GERD's reservoir, which has an estimated capacity of 74 billion m³ of water. That would cut 25 to 33 billion m³ from Egypt's annual quota of Nile water which, in turn, could yield a loss of about 2 million acres of cultivable land [6.3].

Alexandria planned needs from public water supply in 2017 was 2.9×10^6 million m³/day, while water demand for industry was 427×10^6 m³/s, [6.4].

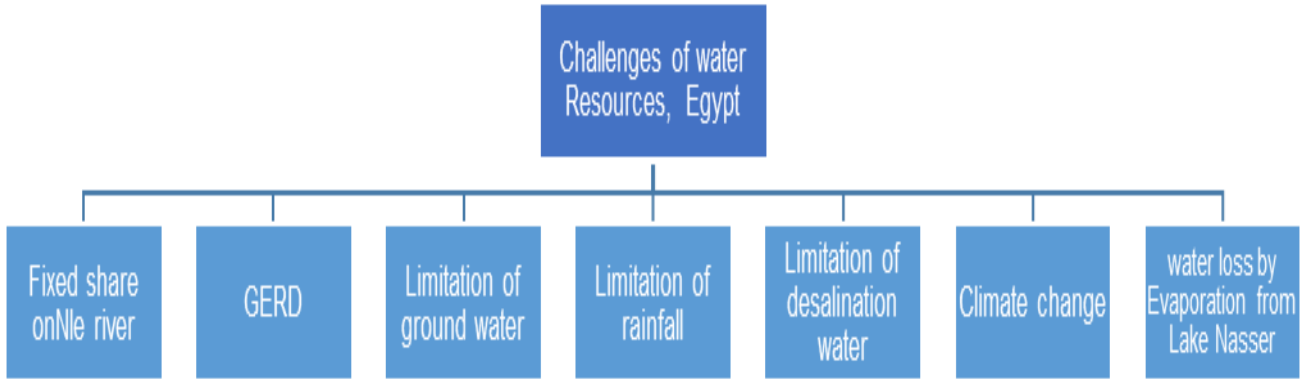


Fig. 6.3: Challenges of water Resources in Egypt [6.1]

Impact of the Grand Ethiopian Renaissance Dam (GERD) [6.5]:

The GERD reservoir's five filling phases started in July 2020 up to September 2024. Temporal changes in the downstream reservoirs are attributed to increased rainfall and flooding. Filling the GERD during drought periods could significantly reduce Egypt's Nile water allocation. The major risk of GERD will happen in drought periods, where the level of water behind the dam will be significantly decreased. Meanwhile the political negotiations failed to find a collaboration between Ethiopia, Sudan, and Egypt is required for an efficient use of the Nile water.

Integrated strategies for water resources in Alexandria, Egypt:

The integrated strategies for water resources in Alexandria includes many approaches either in demand or in supply sides.

1- In the demand-side [6.6]:

Options range from conducting:

- Retrofits at both residential and non-residential properties,
- Tariff re/form,

- Fixing leaks in the network,
- The Regulation of efficient appliances.
- Rationalization of water consumption in all Industrial factories.
- Using new technologies in irrigation systems in agriculture.
- The concept of improving irrigation efficiency in agricultural areas in order to provide more water to the city.

2- In the supply side:

- Seawater desalination for potable use,
- Wastewater recycling for non-potable uses,
- Agricultural drainage water desalination for non-potable uses,
- Local use of groundwater.
- Wastewater recycling for non-potable use in new housing areas.

Governorate	Existing Plants				Future plants		Total design capacity
	Design capacity	Production	Extensions				
			Under constr.	Planned	Under constr.	New planned	
Alexandria	2,414	2,002	0	234	180	0	2,828
Asyut	781	318	0	35	121	45	981
Aswan	227	146	0	26	173	75	501
Beheira	521	492	52	207	138	130	1,048
Beni Suef	356	232	0	43	104	60	563
Cairo	4,122	3,947	0	1,100	400	0	5,622
Dakahlia	861	703	104	86	267	0	1,318
Damietta	335	230	35	86	69	0	526
El Menia	418	286	0	35	475	0	928

Fig. 6.4 Public Water supply ($10^3 \text{ m}^3/\text{day}$) [6.4]

Water

Moaaz Ramadan

Overall goal — vision	Specific goal	Action
Seawater desalination for potable use	Increase potable water supply	Build solar-powered desalination plants.
Wastewater recycling for non-potable uses	Improve irrigation efficiency	Implement modern irrigation technologies in agriculture.
Using new technologies in irrigation systems in agriculture	Reduce water consumption	Implement water-saving regulations, fix leaks, and use efficient appliances.
Construction of huge basin storages to collect storm-water in rainy season	Lower energy costs in desalination	Prioritize solar over wind energy for desalination plants.
Reduce reliance on the Nile and diversify water sources.	Enhance water recycling for non-potable uses	Develop wind-powered and hybrid desalination plants.

It can be indicated that the most cost effective options are those that improve the efficiency of water use in Alexandria, that is, the demand management options, including agricultural efficiency offsets. Many of these water efficiency options have a lower unit cost than the current operating cost of supplying water in Alexandria. This means that there would be a significant net financial benefit to authority, and to the customers, if these low unit cost options were implemented.

The implementation of the water saving options also has the effect of

significantly reducing energy consumption, and therefore reducing greenhouse gas emissions. In addition, there is a significant opportunity for local economic development, in terms of the additional employment that would be generated in the implementation of these options, and also the potential for local manufacture of water saving fittings and fixtures [6.6].

Egypt, like many countries in the Middle East region, faces significant water scarcity issues due to its arid climate and reliance on the Nile River for freshwater. Along with the increase of population which impose much higher demand for suitable water for drinking. As

a result, Egypt has turned to desalination technologies as a vital solution to address the growing demand for freshwater [6.7]& [6.8]. However, despite the importance of desalination plants in Egypt's water strategy, there are several obstacles associated with their implications of those technologies. Thus, these obstacles can be divided into several aspects. Such as, environmental aspect, economic aspect and energy efficiency. [6.9]

By using the conventional methods in desalination plants. Such as, using fossil fuel, will negatively impact the environment due to greenhouse gases (GHGs) emissions. As a result, it will lead to endangering aquatic ecosystems. Fur-

Governorate	Demand in 2000														Estimated demand 2017
	Chemical		Food		Textile		Engineering		Mining		Metal		Total		
	Number of labour (-)	Water use '000 m³/s	Number of labour (-)	Water use '000 m³/s	Number of labour (-)	Water use '000 m³/s	Number of labour (-)	Water use '000 m³/s	Number of labour (-)	Water use '000 m³/s	Number of labour (-)	Water use '000 m³/s	Number of labour (-)	Water use '000 m³/s	
Alexandria	41,473	125,058	33,594	151,842	75,090	36,309	20,265	4,237	10,400	11,138	13,688	19,037	194,510	347,621	426,839
Asyut	1,837	5,539	3,830	17,311	2,236	1,081	691	144	3,386	3,626	16	22	11,996	27,725	56,584
Aswan	3,137	9,459	5,125	23,165	0	0	334	70	1,882	2,016	1,944	2,704	12,422	37,413	69,008
Beheira	1,935	5,835	6,715	30,351	38,922	18,820	303	63	2,542	2,722	238	331	50,655	58,123	74,995
Beni Suef	47	142	2,954	13,352	2,026	980	590	123	1,223	1,310	53	74	6,893	15,980	24,357
Cairo	61,950	186,805	43,746	197,728	31,918	15,434	59,915	12,528	23,917	25,615	55,176	76,737	276,622	514,846	634,707
Dakahlia	5,725	17,263	9,067	40,982	8,507	4,113	2,010	420	1,806	1,934	871	1,211	27,986	65,925	85,457
Damietta	404	1,218	2,063	9,325	4,558	2,204	550	115	388	416	21	29	7,984	13,307	48,249
El Menia	12	36	9,229	41,714	2,126	1,028	773	162	538	576	54	75	12,732	43,591	65,722

Fig. 6.5 Industrial water demand (10³ m³/s) [6.4]

thermore, the operational costs of desalination are also high, mainly because of energy consumption, maintenance and ongoing operational costs. [6.10]&[6.11] Since desalination plants powered by renewable energy can provide sustainable solutions for the lack of freshwater issue, along with reducing the cost within the desalination industry, which considered as a major concern. Hence, a case study was conducted in Egypt between two renewable energy options that are commonly used in desalination industry are solar and wind energies. In order, to determine the best of these options to power up the desalination plants with least cost. According to the author's findings, employing solar or wind energy results in lower electricity bills than using conventional energy sources. In addition, by applying the techno-economic analysis, the simulation findings shows that the proposed desalination plant powered by solar energy can produce water at a levelized energy cost of 3.95 cent/kWh. While the

desalination plants powered by wind energy can produce water at a levelized energy cost of 9.5 cent/kWh. Additionally, a meter cubic of filtered water costs 0.08 when using a solar photovoltaic system and 0.19 when utilizing a wind turbine system. Hence, the ideal option for this case study is a solar photovoltaic desalination system [6.12]. Another potential answer to the issue may be located in Marrakesh, Morocco's tourist center, where an antiquated technology is being revitalized due to ample wind and sunlight. In Morocco, power from renewable sources like wind and solar has grown. Today it stands at about 20% according to the Moroccan Agency for Sustainable Energy. According to GWI data, the cost of desalination plant water has decreased by up to 60% over the last ten years, to roughly 37 cents per cubic meter. According to the International Energy Agency, the price of renewable energy has decreased by almost two-thirds in recent years. Refer to Figure 6.5 [6.13]

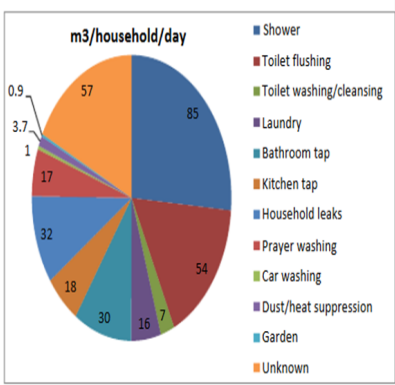


Figure 6.7 Breakdown of household end use in Alexandria (2010) [6.7]

Coxabengoa, a Spanish water and energy company, runs the Agadir desalination plant, that produce 275,000 cubic meters of water per day, of which 150,000 cubic meters are utilized for drinking water, which is sufficient to meet a million people's basic daily needs. The remaining water is used for irrigation. Moreover, A larger 800,000 cubic meter desalination plant is being built near Casablanca, along with several expansions in Morocco in order to achieve their vision [6.13]&[6.14].

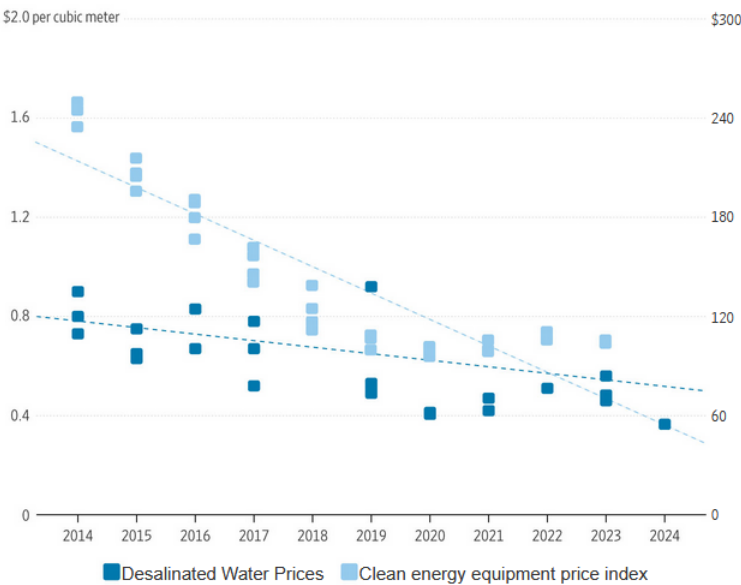


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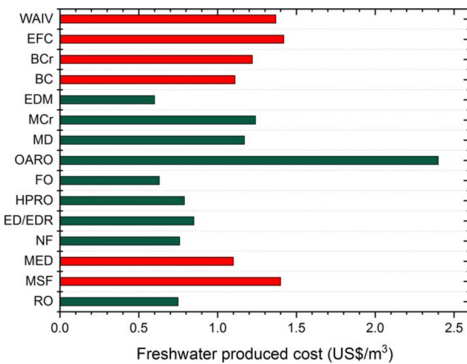


Figure 6.8 Freshwater Produced cost (USD/m³) [6.10]

Circularity

Akshay Maru veettil

Overview (2016)

Total urban solid waste generated	5000 tons [7.2]
Urban solid waste generated per capita	339.45kg per person per year 0.7kg per person per day
Urban solid waste collected	80%
Recycling rate	21%
Collection and Transport	2000 tons per day

Developments in Circularity

Alexandria is actively adopting to the circular economy much like the rest of the Egypt aimed to encouraging environmentally conscious and switching the methods towards circular economy value chains project. The national focus has been on promoting sustainable waste managements that leads to recycling and resource recovery by implementing waste management laws (Law No. 202 of 2020). Implementation of this law made an increase in recycling by



Fig. 7.1: Waste separation bins based on different colours.[7.2]

5% by this time and rest is either burned or dumped in a proper manner. However initiatives like the law implementations and the National solid waste management program which is co financed by the EU aims to create a climate sensitive waste management inorder to change the current linear waste management method into the circular economy.[7.1] The changes happening is not only in the technical or legislation, it also takes significant efforts in training, policy implementations and encouraging private sector involvements in green practices.

Waste Management

Waste management in Alexandria is an combination of multiple techniques such as Integrated waste management systems (ISWM), recycling initiatives composting and organic waste management, Land fill management and upgradations, public engagement and education. Integrated waste management systems is a method implemented for management of the waste produced in corporation with other facilities available in the city. The system includes transfer stations which serves as the points where the wastes are

Share of waste disposal (2021)

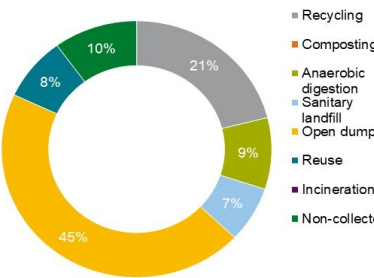


Fig. 7.2: graph [7.13]

collected and stored for the segregations and dumping into the respective sites as a measure for reducing environmental impacts. Organic waste management and composting includes collection, breakdown and usage of composite to recycle it to make fertilizers.[7.4] Land fill management are systems to mitigate the environmental impact by applying linear and leachate systems which prevents the contaminations from mixing with ground-water along with gas collection systems to reduce greenhouse gas emissions and rehabilitation projects.

Waste Recycling

The waste recycling rate in Alexandria currently stands in 21% (Fig 7.1) which is extremely low compared to other cities in Egypt. Increasing population in the city is eventually putting the current recycling methods to failure along with creating high tension on the authorities to achieve higher levels of circularity, however implementing new strategies and methods by the current authorities is exploring the region necessities and bringing up new methods and innovative ideas to increase the circularity level.[7.3]

Challenges and Measures

Alexandria being under the changes towards the circular economy undertakes a lot of challenges which begins from collection and recycling rates which is facing difficulties due to the uneven recycling rates, in urban areas the coverage is around 40% to 85% whereas in rural areas is 0 to 35%.[7.5] Another challenge they face is the lack of institutional and regulatory frameworks which delays the supports for the circular economy . Another challenge for an circularity in waste management is the public awareness and engagement in recycling and sustainable waste management practices. Another challenge faced by the authorities is reduced recycling rates which makes it difficult for improvising and reducing waste emissions in the form of recyclable materials.[7.6] Economic and regulatory challenges is always a hurdle for the cities that are developing due to financial situations and regulatory constraints which slows down the process. When it comes to measures taken for

improving the current scenarios, implementation of legal reforms as mentioned above in developments which aims in providing structure for the circular economy principles across the cities. Efforts have been taken into the infrastructure developments despite of the challenges faced to handle waste in most effective ways with the supports from the private sector and NGOs.[7.7] Despite these challenges Alexandria is improving it's waste management and promote sustainability through principles of circular economy. Furthermore the collaboration with the European union for circular economy transition the city has implemented some of their methods similar to EU standards prevention methods are reuse, recycle and reduction of landfill use which supports to gradually increase recycling rates over the period of time along with following of EPR extended producer responsibility designing them to process and use of secondary raw materials. These initiatives shows the city trying to improve the environmental

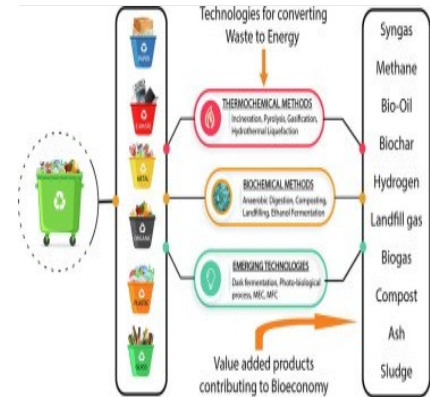


Fig. 7.4: Conversion of waste to energy chart [7.6]

goals. EU and other entities provide supports and financial aids for the city and country in maintaining the above mentioned challenges. The developments in other countries in the respective fields of waste management is undergoing efforts through the helps of funding and partnerships with the private sectors and people which tailors into developing the innovations towards local challenges and development of recycling markets supporting the pricing and competitiveness of recycled materials [7.12] EU and other NGOs also helps the city by providing targeted funding and the b using the programs such as cohesion fund and regiona development funds to assists regions in developing the necessary infrastructure.[7.16].

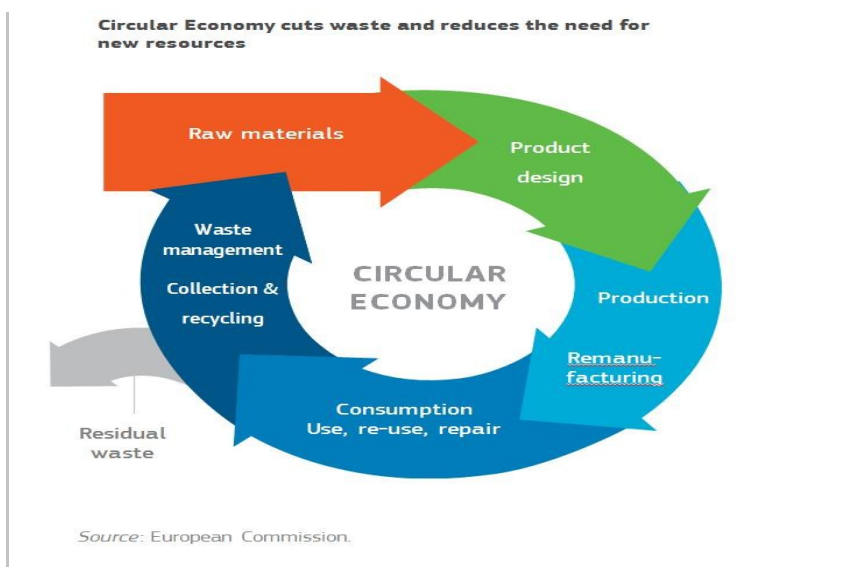


Fig. 7.3: Circular economy flow chart [7.16]

Circularity

Akshay Maru Veettil

Overall goal — vision	Specific goal	Action
Transforming waste management which focusing on stability within circular economy.	Maximizing resource efficiency	Upgrading recycling facilities and investing in research and development.
Environmental protection and sustainability	Waste to energy developments	Construction of waste to energy plants
Economic development through circular economy practices.	Leveraging public private partnership	Foster collaborations and engage in technology transfers.

Linear to Circular developments

Changing from linear circularity to circular economy is having their own strategies which enhances the air and water quality along with human health. The major focus for the stability of the waste management by the infrastructure development with construction of waste to energy plants involving planning , financing and non renewable energy resources and upgrading recycling facilities by modernizing recycling infrastructure to handle higher volumes of waste more efficiently is crucial and process recyclable materials more accurately enhancing the overall recycling rate. Furthermore legislative and policy support brings enact supportive policies which develop projects for supporting and promoting the waste management which also includes tax reductions for

companies investing in recycling technologies or subsidies for achieving waste reduction.[7.7] Strengthening environmental regulations to ensure sticter enforcement of disposal and treatment standards. Law enforcement also sets limits on landfill use and mandating the segregation of waste at the source.

Enhancing recycling infrastructure

Enhancing recycling infrastructure in Alexandria involves several strategic steps that can help the city manage it's waste more effectively and sustainability. Upgrading facility technology is a prime factor for enhancing the recycling infrastructure which includes advanced sorting technology. Incorporate state of the art sorting machines that can segregate recyclables such as plastics, metals paper and glass which will make the reclying. Introduction of robotic sorting systems such as application of artificial intelligence and robotics which eventually end up in cost reduction. [7.10] Expansion of collecting and sorting facilities by decentralizing the collecting centres and mobile recycling units by deploying units in neighbourhoods to collect recyclables. Leverages using Public private partnerships by engaging private sector expertises by partnering with private companies and people companies that specialize in waste management. Innovative financing models by such as green bonds, private equity or

public grants that can provide the necessary fund.

Waste to Energy facilities

Waste to energy facilities are an integral part of modern managements in waste solutions converting municipal and industrial waste into energy in the form of electricity or heat. Alexandria's method of Waste to energy includes incineration, gasification, anaerobic digestion Landfill gas capture. Most effective and cost efficient method is incineration where waste burned at high temperature and the heat generated is used to produce steam that drives turbines to generate electricity. Another effective method which can be used landfill gas capture which uses methane gas emitted from decomposing organic waste in landfills significantly reducing the greenhouse gas emissions from the landfill. Benefits of WtE facilities is energy production help reduction in fossil fuels and reduction in waste volume by incineration reduces the volume of waste that needs to be landfilled by upto 90% significantly extending the life of existing landfills. [7.9]

The cycle is completed when consumers purchase products made from recycled materials. This demand encourages more recycling and the production of goods from recycled materials.



Fig. 7.5: Local solid waste disposal facilities [7.17]

Promoting secondhand goods and value recovery

Promoting secondhand goods and enhancing value recovery are crucial strategies for making a sustainable economy and reducing environmental impact. Strategies for promoting secondhand goods includes awareness and education about the benefits of buying and using secondhand goods made from recycling which will eventually reduce waste, save money and minimize their carbon footprint. Quality and trust among consumers is another factor to be considered by the authorities for providing the quality standards and certifications for the platforms. Another method is incentivization is the sales or donations through financial incentives such as tax deductions, vouchers or recycling programs. Enhancing value recovery includes repair and refurbishing programs which also creates job opportunities in the repair and refurbishment sector.

Interdependencies

In Alexandria several interdependent projects aimed at promoting circularity in waste management with the initiatives in the national solid waste management program, the program is co financed by the EU and implemented by GIZ.[7.11] The program seeks to transform waste management systems into a more climate sensitive and circular economy oriented model. Additionally, private sector involvement has been advancing waste management practices. This application and other similar initiatives are part of a broader move towards improving the economic value of waste and integrating informal waste collectors into formal waste management systems. Integration of multi-sector and advanced technologies along with community engagement are to the key to the future success of these initiatives. [7.13]



Fig. 7.7: Energy cycle [7.19]

Conclusion

Alexandria's commitment to circularity in waste management under the current situations can clearly shows the improvisation on how urban areas can transfer waste management into sustainable development and making opportunities. By integrating the technological advancements and strengthening policy works and engaging the communities the city is moving towards a more sustainable and economical stability in waste management.



Fig. 7.6: Waste transportation in Alexandria [7.11]

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Alexandria, Egypt, is facing serious waste management and environmental challenges that threaten both public health and the city's future sustainability. Every day, large amounts of solid waste are generated, but collection and recycling efforts are not as effective as they should be. Many areas still rely on open dumping and non-sanitary landfills, which leads to pollution and increases the risk of diseases. Without better waste disposal and recycling systems, the city will continue to struggle with growing waste-related problems.

Beyond waste management, Alexandria is experiencing rapid urban growth, which has led to overcrowding and the expansion of informal settlements. Many of these settlements lack proper infrastructure, such as sewage systems, clean water, and waste collection services. As a result, pollution and poor living conditions are becoming more severe. The city's development has often been unplanned, leading to chaotic zoning and inefficient land use. If these issues are not addressed, the quality of life in Alexandria will continue to decline.

Another major challenge is climate change. Rising sea levels pose a serious threat to Alexandria's coastal areas, with some neighborhoods already experiencing flooding during storms. Without proper coastal protection measures, parts of the city could become permanently submerged in the coming decades. Additionally, air pollution is becoming a growing concern due to increased traffic and industrial activities. A lack of green spaces worsens air quality, making it harder for residents to enjoy a healthy and comfortable urban environment.

To address these challenges, Alexandria needs a well-planned waste management system that prioritizes reducing waste, improving recycling, and using modern disposal methods. Encouraging businesses and residents to participate in recycling programs can help reduce the amount of waste that ends up in landfills. Investing in better waste treatment facilities will also be crucial to ensuring that garbage is processed in an environmentally friendly way.

At the same time, urban planning must improve to provide better living conditions for residents. Expanding green spaces and enforcing stricter building regulations can help create a more livable city. Upgrading public transportation, such as introducing more buses and a reliable metro system, can reduce traffic congestion and lower air pollution levels. Better zoning laws can ensure that residential, commercial, and industrial areas are properly separated, reducing environmental hazards.

Water conservation and renewable energy should also be a priority. Alexandria needs to invest in solar and wind energy to reduce its dependence on fossil fuels, which contribute to pollution and climate change. Coastal protection projects, such as seawalls and flood barriers, must be expanded to safeguard vulnerable neighborhoods from rising sea levels. Additionally, improving water treatment facilities can help ensure that residents have access to clean drinking water despite increasing demand and pollution risks.

Ultimately, solving these problems requires cooperation between the government, private sector, and local communities. Strong policies and investments in sustainability can help Alexandria

Sustainability rating

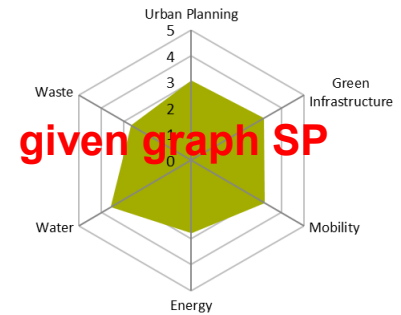


Fig. 8.1: Sustainable rating of Alexandria [1.4]

transition to a cleaner and more environmentally friendly city. If all stakeholders work together to improve waste management, urban planning, and climate resilience, Alexandria can secure a better future for its residents and protect its rich history and coastal beauty.

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